# AI in Operations and Supply Chain

Week 5 · From forecast to decision



# Some Definitions (Cluttered, I know :))

**Operations:** How an organization turns inputs into the goods or services it sells.

- Inputs include materials, people, equipment, information and money;
- Outputs include a loaf of bread, a phone contract or a hospital treatment.

**Operations management:** designs, runs and improves those processes, usually judged on cost, quality, speed, flexibility and reliability.

**Supply Chain:** widens the view from one firm to the whole chain of organizations that brings a product to the final customer

→ Raw-material suppliers, manufacturers, distributors, retailers and returns.

Supply chain management coordinates three flows across that network:

- **Materials moving downstream:** goods and inventory.
- **Information moving both ways:** orders, forecasts, sales data.
- **Money moving upstream:** payments.

# A forecast is an input, not an answer

**1** Forecast ≠ decision

**2** Newsvendor and the critical ratio

**3** What a better forecast is worth

**4** Automation and adaptive planning

**Lab** Order quantities in Jupyter

90 min

**By the end of today you can**

- Explain why the same forecast should lead to different orders for different products
- Compute an order quantity from a forecast using the critical ratio
- Put a money value on a more accurate forecast and use it as a ceiling price
- Decide when automated document processing needs a human in the loop

01

# A forecast is not a decision

# Two ways to get it wrong

## Too much $4.3 bn

of unsold clothes reported by H&M in 2018: cash tied up, then marked down or written off.

**Summary:** perating profit for December 2017 to February 2018 fell 62%, H&M's weakest first quarter in 16 years and below analysts' expectations. Unsold inventory had risen to about **17.6%** of annual sales, a level analysts called "worrying" because it kept piling up.

**Weather:** H&M blamed it partly on the weather. An unusually warm January in Europe was followed by a cold snap in February, so the winter and spring ranges both arrived at the wrong time.

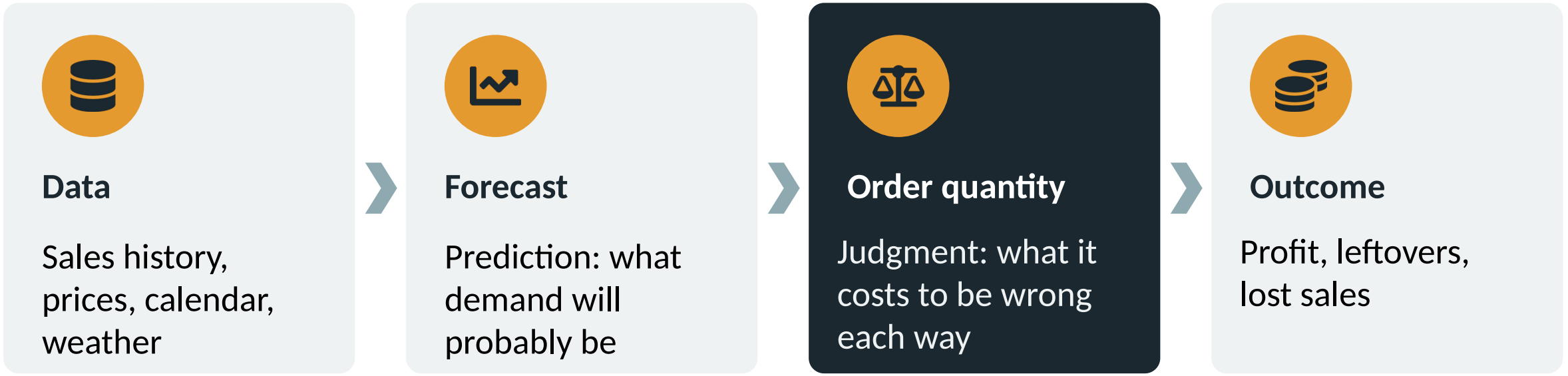
**Underlying problem:** it went beyond the weather. Same-store sales had fallen for ten quarters in a row, and the share price was down 56% over three years.

## Too little 10:00

A bakery that never throws bread away but sells out every morning: every customer after ten is a lost margin

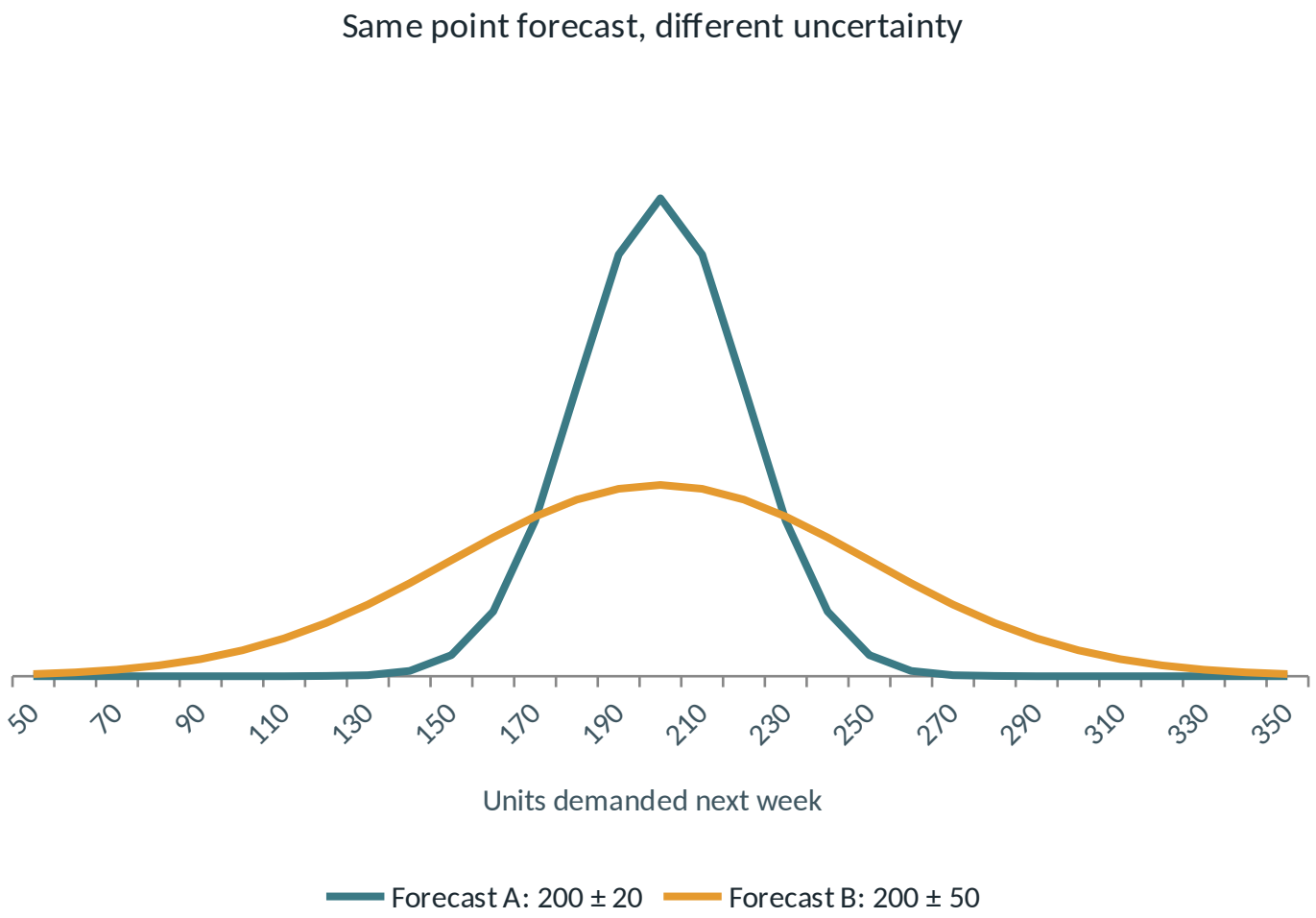
***Both are decision failures as much as forecast failures***

# Prediction is cheap; judgment decides the outcome




Agrawal, Gans & Goldfarb: AI makes prediction cheaper, so the value moves to the judgment that uses it (recall week 1)

# The decision needs the spread, not just the mean




**200**

units is the point forecast for both

- The point forecast is the mean of a distribution
- How much to order depends on σ, the spread
- Ask forecast providers for intervals or quantiles, not one number

# Forecast accuracy is not business value

| | Forecast A | Forecast B |
|---|---|---|
| **Mean absolute error** | 30 units | 30 units |
| **Bias** | 0 (errors cancel) | -25 (always too low) |
| **High-margin product** | Small, balanced misses | **Repeated stock-outs** |

**MAE (mean absolute error):** the average size of a miss, ignoring direction: average of |e|. “On average we’re off by 30 units.” Recall MSE*

**MAPE (mean absolute percentage error):** the average of |e| ÷ actual demand, as a percentage. “On average we’re off by 12%.” It has no units, so managers like it for comparing products and stores.

**Bias (mean error (forecast - demand)):** the average of e with its sign. It answers a different question: not “how big are the misses?” but “do we miss in one direction?” Under this convention, negative = forecast too low, but some forecasters flip the sign!



### MAE and MAPE treat over and under equally

The business rarely does: a lost sale and a leftover cost different amounts



### Bias pushes every order the same way

A small, steady bias can cost more than a larger random error



### Judge a forecast by the profit of the decisions it drives

That is the test we use in the lab

# 02

# Asymmetric costs and the newsvendor

# The newsvendor: order before demand is known

**1. Order q units now**

At unit cost c

**2. Demand D is revealed**

Uncertain; we have a forecast distribution

**3a. If D < q: leftovers**

Unsold units go for salvage value v (or zero)

**3b. If D > q: lost sales**

Each missed customer costs the margin p − c

**Where it shows up**

- Fresh food: bread, fish, dairy, flowers
- Fashion: one production run per season
- Event catering and hotel food and beverage
- Seasonal goods: calendars, school supplies, Christmas stock
- Vaccines and spare parts ordered once per cycle

# Two mistakes, two prices

### Under-stocking cost

$$Cu = p - c$$

The margin you lose on each sale you could have made

***Can add: lost goodwill, customers who switch store***

### Over-stocking cost

$$Co = c - v$$

The money you lose on each unit you could not sell

***Can add: disposal, holding and markdown costs***

**p = selling price · c = unit cost · v = salvage value per leftover unit**

# Should we stock one more unit?

Unit q + 1 sells if demand exceeds q

**Stock it if $P(D > q) \cdot Cu \geq P(D \leq q) \cdot Co$**

*expected gain if it sells ≥ expected loss if it does not*

Rearranged: keep adding units while

$$P(D \leq q) \leq Cu / (Cu + Co)$$

Stop when the chance of selling the next unit is too low to pay for it

**Same logic as week 3**

- Churn: call a customer if p × benefit beats (1 − p) × cost
- That gave a break-even probability
- The critical ratio is the break-even probability of the last unit you stock

# The critical ratio sets the order

$$CR = Cu / (Cu + Co)$$

The share of demand outcomes you want to cover

$$q^* = \mu + z \cdot \sigma$$

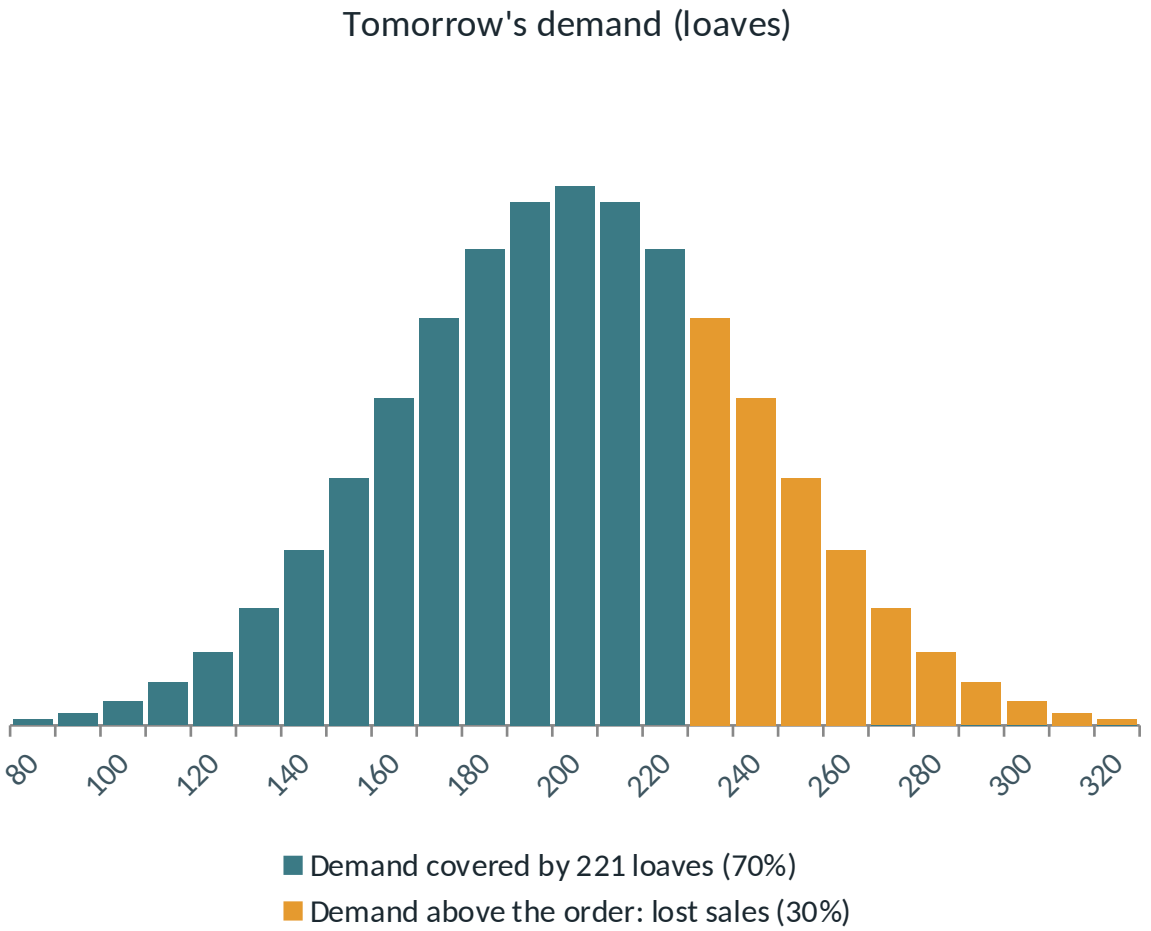
- μ, σ: mean and spread of the demand forecast
- z: the CR-quantile of the standard normal, =NORM.S.INV(CR) in Excel

| CR | z | Order vs. forecast |
|---|---|---|
| 0.20 | −0.84 | well below |
| 0.50 | 0.00 | exactly the forecast |
| 0.70 | 0.52 | a little above |
| 0.80 | 0.84 | above |
| 0.90 | 1.28 | well above |
| 0.95 | 1.64 | far above |

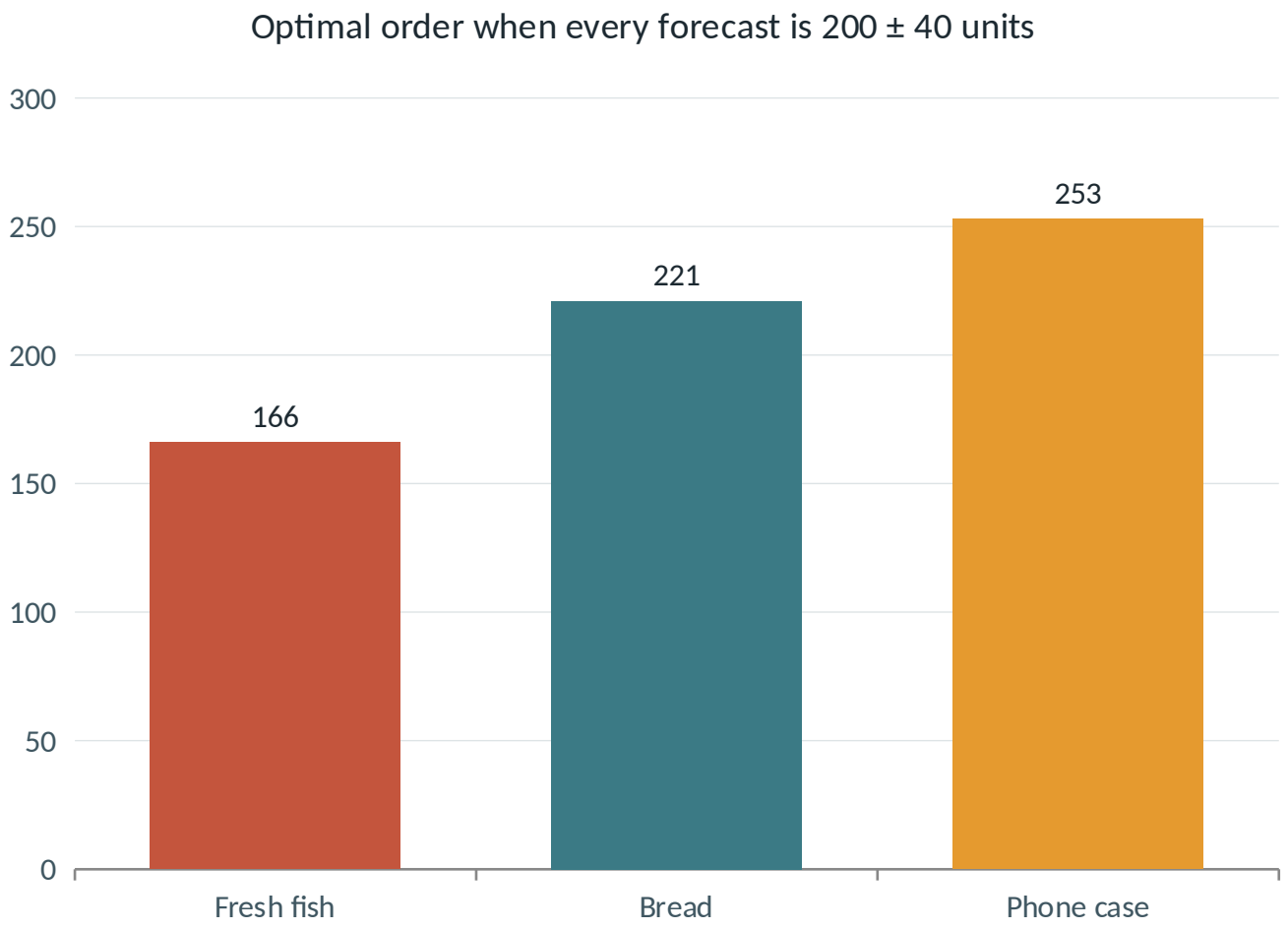
*Assumes normally distributed forecast errors*

# Worked example: bread in a bakery

| | |
|---|---|
| **Forecast** | μ = 200 loaves, σ = 40 |
| **Prices** | c = €0.30, p = €1.00, v = €0 |
| **Costs** | Cu = €0.70, Co = €0.30 |
| **Critical ratio** | 0.70 / 1.00 = 0.70 |
| **z** | NORM.S.INV(0.70) = 0.52 |
| **Order** | **200 + 0.52 × 40 ≈ 221 loaves** |



# Same forecast, different decisions




**Fresh fish**

c €8, p €10, v €0

**CR 0.20 → order below forecast**

**Bread**

c €0.30, p €1, v €0

**CR 0.70 → a little above**

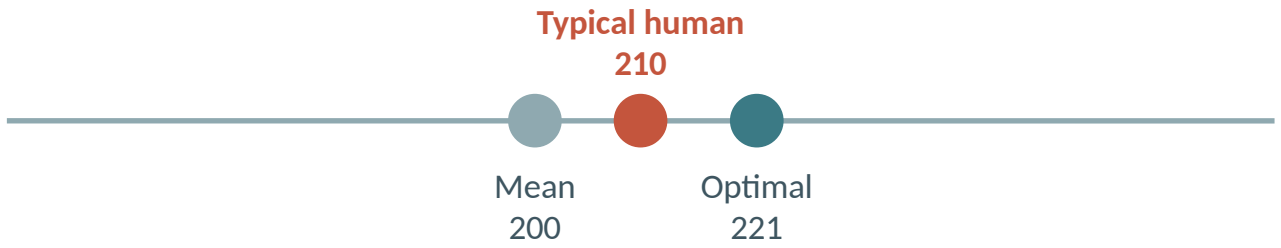
**Phone case**

c €2, p €12, v €1

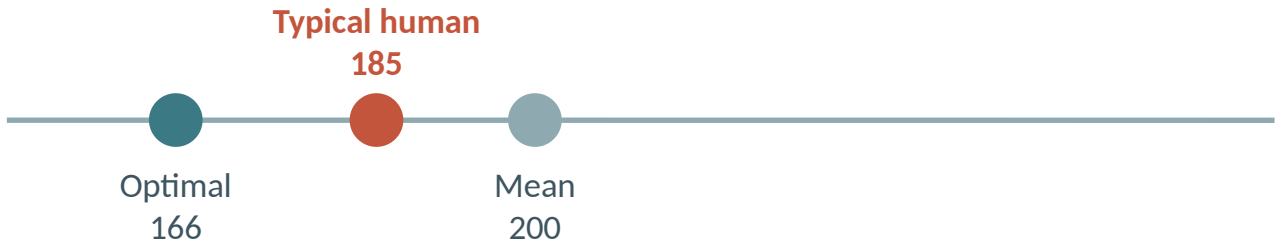
**CR 0.91 → well above**

# People know the rule but still order near the mean

**High-margin product (CR 0.70)**




**Low-margin product (CR 0.20)**




*Human positions are illustrative of the pattern, not measured values*

**Pull to center**

- Schweitzer & Cachon (2000): in lab experiments, people's orders fall between the mean and the optimum
- They learn slowly, even with feedback
- **So automate the decision rule, not only the forecast**

# 03

# What a better forecast is worth, in money

# The cost of uncertainty grows with σ




At the optimal order

**Expected cost = (Cu + Co) · φ(z) · σ**

- φ(z) is the normal density at z; it depends only on the costs
- So the cost is a straight line in σ: cut σ by 30%, cut the cost by 30%
- A perfect forecast (σ = 0) makes the cost zero

# Pricing an ML forecast for the bakery

**−30%**

**forecast error**

σ falls from 40 to 28 loaves

**€4.17**

**saved per day**

€13.91 → €9.74 mismatch cost

**≈ €1,500**

**per product per year**

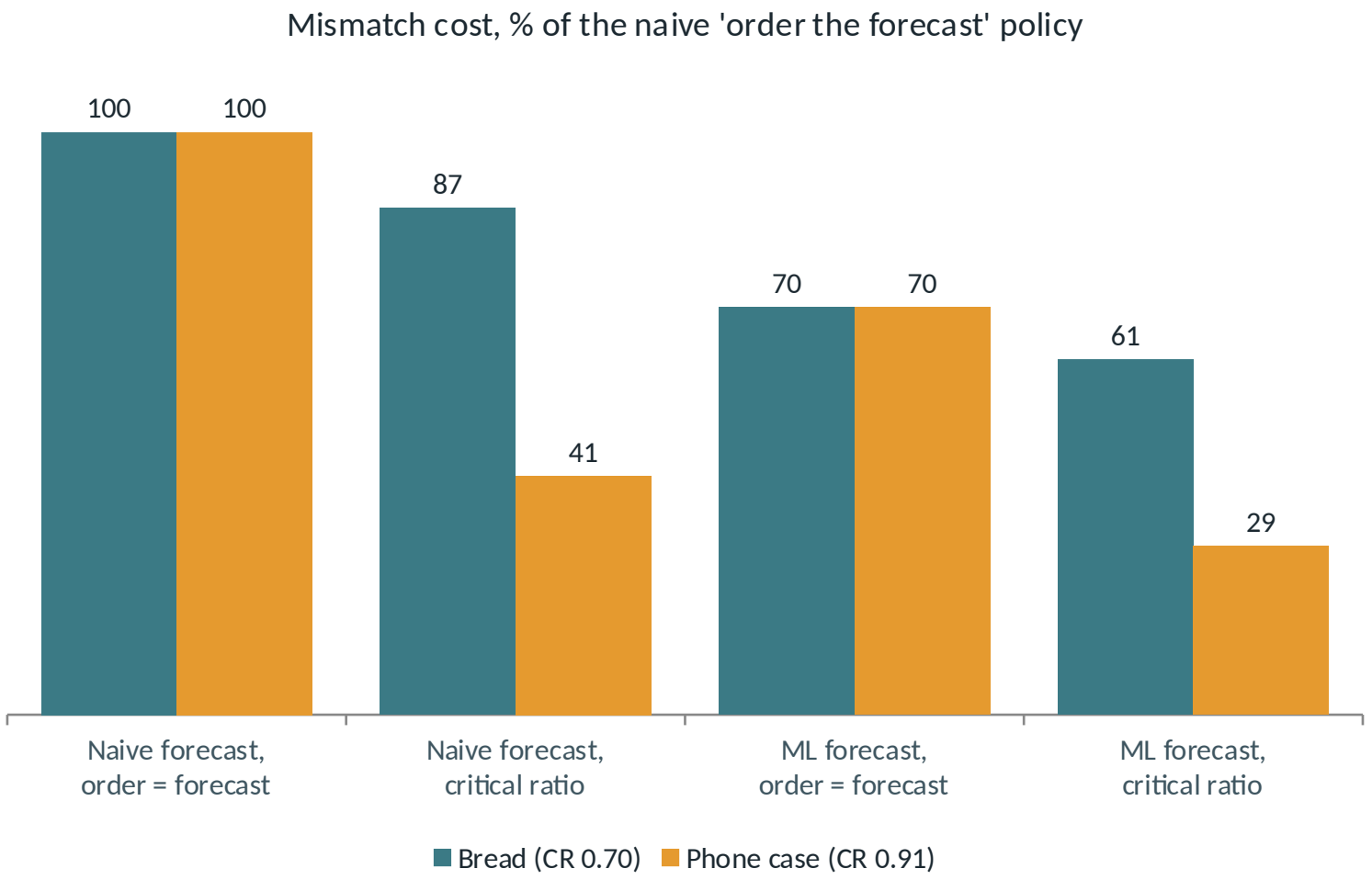
€4.17 × 365 days



**The saving is a ceiling price**

- Multiply by products and stores: 200 products ≈ €300,000 a year, before costs
- Subtract data, integration, monitoring and staff time: the total cost of ownership from week 7
- If the vendor costs more than the saving, keep the naive forecast and use the critical ratio

# The right rule can beat the better model




**Two separate gains**

**Decision gain**
Use the critical ratio. Almost free.

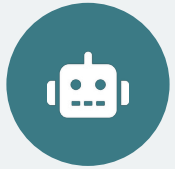
**Forecast gain**
Buy or build a better model. Costs money.

*The more asymmetric the costs, the more the rule matters*

# 04

# Automation and adaptive planning

# Rule-based RPA vs. document AI



## Rule-based RPA

*Software robot that clicks and types like a clerk, following fixed rules*

- Cheap, fast to build, fully auditable
- Works on stable screens and fixed formats
- Breaks when a layout or field changes
- Best for high-volume, unchanging steps



## Document AI

*Models (including LLMs) that read invoices, orders and delivery notes*

- Handles many suppliers and layouts
- Returns structured fields plus a confidence
- Makes probabilistic errors: needs evaluation
- Recall week 4: JSON output, scoring on labels

**In practice they are combined: document AI reads the document, RPA keys the result into the ERP**

*** Robotic process automation (RPA)**

# When should a person check the machine (LLM)?

Send to a human if

$$(1 - p) \cdot C_error > C_review$$

**p = model's confidence that the extraction is correct**

Example: a wrong invoice amount costs €50 to fix later; a review costs €2

**Review everything below 96% confidence**
**0.04 x 50 = 2**

**Confidence scale**




- Same expected-value logic as the critical ratio
- Only works if the confidence is calibrated: check it on a labelled set (week 4)
- Raising C_error (payments, safety) pushes more cases to people
- Track the review rate and the error rate after go-live

# Adaptive planning: decide more often, closer to demand

### Bullwhip effect

Small swings in shop demand grow into large swings at suppliers when each stage orders in batches on its own forecast

*Lee, Padmanabhan & Whang (1997)*

### Shorter cycles

Re-forecasting nearer the sale shrinks σ, and the cost falls with it. Zara's stores reorder twice a week

*Ferdows, Lewis & Machuca (2004)*

### Real-time signals

Point-of-sale data, weather and local events feed the next order automatically instead of the monthly plan

*Where ML adds most value*

***Often a shorter decision cycle is worth more than a better model***

# Key takeaways

1. A forecast is a distribution; the order comes from costs
2. Critical ratio = Cu / (Cu + Co) is week 3's break-even probability
3. Price a better forecast by the money it saves, not by its MAPE
4. Put people where the expected cost of a machine error exceeds the review cost

**Next: the lab. Choose order quantities for a retailer from naive and ML forecasts, simulate profit, and price the better forecast**

Reading: Cachon & Terwiesch, Matching Supply with Demand (newsvendor chapters) · Fisher (1997), HBR · Schweitzer & Cachon (2000) · Hyndman & Athanasopoulos, FPP ch. 1 (optional, but highly recommended if you want to understand forecasting)